AC444
VALUATION
AND SECURITES ANALYSIS

Week 8 Session 1
FCF-Based Valuations
Structure of Valuations

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TOPICS DISCUSSED

- Prospective Analysis: Forecasting;
- DCF-Based Valuations;
- A Practical Approach to Valuation.

PROSPECTIVE ANALYSIS: FORECASTING

FORECASTING IN THE EXPLICIT FORECASTING PERIOD

- Key forecasts (Income statements):
  - ✓ Revenue growth;
  - ✓ Gross margins (GM);
  - ✓ Different types of operating expenses;
  - ✓ Tax expenses;
  - ✓ Net financial expenses.

FORECASTING IN THE EXPLICIT FORECASTING PERIOD

- Key forecasts (Balance Sheet statements):
  - ✓ Book values of non-current operating assets;
  - ✓ Book values of net operating working capital components;
  - ✓ Book values of net financial obligations.

- These assumptions will generate forecasts of book values of equity and net dividends (ND also referred to as FCFE).

FORECASTING IN THE EXPLICIT FORECASTING PERIOD

- There are 2 different approaches:
  - ✓ a bottom-up approach;
  - ✓ a top-down approach.

PROSPECTIVE ANALYSIS
Application to the Schneider- Square D Case-Study

PROSPECTIVE ANALYSIS

Application to the Schneider- Square D Case-Study

- The Chairman and CEO of Schneider hired Lazard Freres to advise on the potential acquisition of Square D;

PROSPECTIVE ANALYSIS
Application to the Schneider- Square D Case-Study

- Schneider competes in the electrical and industrial automation industry;

- Core activities:
  - ✓ Electrical power equipment;
  - ✓ Industrial controls and automation equipment;
  - ✓ Nuclear power equipment and propulsion systems;
  - ✓ Electrical engineering and contracting.

PROSPECTIVE ANALYSIS

Application to the Schneider- Square D Case-Study

- Historically, the electrical equipment industry has been segmented by country or by region. Barriers to entry in different markets have been determined by:
  - Differences in standards across countries and regions;
  - Costs of R&D for new products and costs of translating technologies for different regional standards;
  - Proprietary distribution networks (Square D);
- New trends in the industry towards increased globalization and concentration:
  - Product standards are becoming common across regions;
  - R&D costs continue to grow increasing the benefits of economies of scale.

PROSPECTIVE ANALYSIS
Application to the Schneider- Square D Case-Study

- Schneider and Square D appear to be a good match:
  - Geographical perspective: Schneider is a leader in Europe whereas Square D is a major player in the US market;
  - Firms' strategies and products are highly complementary: Schneider appears to be especially strong in industrial distribution whereas Square D in residential distribution.

- Synergies between Schneider and Square D:
  - Rationalization of R&D efforts and benefits from sharing existing technologies;
  - Access to larger distribution channels for both companies (licensing agreement?);
  - Rationalization of manufacturing capabilities;
  - Benefits from cross-selling products (Square D network).

PROSPECTIVE ANALYSIS

Application to the Schneider- Square D Case-Study

- Lazard Freres calculate the stand-alone value of Square D using the PVFCFE valuation method and the set of key assumptions found on one of the next slides;

- To calculate the value of Square D to Schneider, they add to the stand-alone value of Square D the present value of the expected synergies ($60m per year).

PROSPECTIVE ANALYSIS
Application to the Schneider- Square D Case-Study

- Comparable Financial Data on Square D and Competitors:

	Amp Inc.	Cooper Industries	Emerson Electric	WW Grainger	Square D
Sales ($m)	3043.6	6222.2	7573.4	1935.2	1653.3
NP ($m)	287.1	361.4	613.2	126.8	120.7
Sales Growth (CAGR Yr-3 to -1)	6.8%	20.9%	6.7%	12.3%	5.1%
EBIT Growth (CAGR Yr-3 to -1)	-4.1%	30.2%	8.3%	7.0%	-0.2%
EBIT Margin	16.0%	13.5%	14.6%	10.5%	10.8%
PE Ratio	17.1	14.4	13.9	14.7	9.4
BVNFO/BVNOA	1%	38.6%	11.3%	0.9%	24.2%

PROSPECTIVE ANALYSIS
Application to the Schneider- Square D Case-Study

- Past ratios:

(%)	Yr -6	Yr -5	Yr -4	Yr -3	Yr -2	Yr -1
Sales growth		4.2	4.4	12.5	6.7	3.4
EBIT/Sales	17.4	16.8	15.9	13.1	12.0	12.9
NWC/Sales	16.5	16.0	14.5	11.9	12.1	23.6
Dep expense/Sales	2.7	3.0	3.2	3.0	3.1	3.6
CAPEX/Sales	5.1	5.6	2.7	4.8	5.0	5.7

PROSPECTIVE ANALYSIS

Application to the Schneider- Square D Case-Study

- Explanations:

  - ✓ Square D's performance declined from 17.4% in Year -6 to 12.0% in Year -2 (EBIT/Sales);
  - ✓ Despite the current recession, there is a modest upturn in Year -1. Is this an indication that the firm's restructuring efforts are paying off?
  - ✓ Net Working Capital/Sales steadily declined until Year -2;
  - ✓ The subsequent increase is due to a large build-up in cash;
  - ✓ Capital Expenditure/Sales declined dramatically in Year -4 as the firm responded to the deterioration in its business by cutting back on new investments.

PROSPECTIVE ANALYSIS
Application to the Schneider- Square D Case-Study

- Key assumptions used in the prospective analysis to value Square D on a stand-alone basis:
  - ✓ Sales Growth of 3.5% next year and 7% per year thereafter;
  - ✓ EBIT/Sales of 15 to 16%;
  - ✓ Net Working Capital/Sales at a minimum of 11 to 13%;
  - ✓ Projected Capital Expenditure/Sales of 5%;
  - ✓ Depreciation Expense/Sales at 4% between next year and 7 years from now and 4.3% thereafter.
- The cost of equity capital is assumed to be equal to 16% throughout the valuation.

PROSPECTIVE ANALYSIS

Application to the Schneider- Square D Case-Study

- Please take time to work out in your assigned groups;
- How reasonable is the set of assumptions introduced by Lazard Freres?

Exhibit 1 Pro Forma Financial Statements and Valuation Using Lazard Assumptions (continued)

	1990		1991	1992	1993	1994	1995	1996	1997	1998
BALANCE SHEET										
Cash	244,933	Grows at rate of sales	253,506	271,251	290,239	310,555	332,294	355,555	380,444	407,075
Net Working Capital	261,153	See WC assumption	205,342	219,716	235,096	251,553	269,162	288,003	308,163	329,735
Leveraged Leases	137,182	Grows at rate of sales	141,983	151,922	162,557	173,936	186,111	199,139	213,079	227,994
PPE (gross)	799,367	See Capt exp. assumption	884,926	976,475	1,074,431	1,179,245	1,291,396	1,411,397	1,539,799	1,677,188
Accumulated Depn	349,265	See depreciation exp	417,712	490,951	569,317	653,168	742,888	838,889	941,610	1,059,765
Net PPE	450,102	Plug	467,214	485,524	505,115	526,078	548,508	572,508	598,188	617,423
Other Assets	145,804	Grows at rate of sales (w/ sale discont. ops. in 1991)	150,907	161,471	172,774	184,868	197,808	211,655	226,471	242,324
Total Assets	1,239,174		1,218,952	1,289,884	1,365,780	1,446,989	1,533,883	1,626,860	1,726,345	1,824,550
Liabilities & Equity										
Interest Bearing Debt	383,758									
Unearned ESOP Compensation	(120,400)									
Net Debt	263,358									
Deferred Taxes	210,080	Grows at rate of sales	272,576	291,656	312,072	333,917	357,291	382,301	409,062	437,697
Other Liabilities	26,633	Grows at rate of sales	217,433	232,653	248,939	266,365	285,010	304,961	326,308	349,150
Minority Interest	10,941	Grows at rate of sales	27,565	29,495	31,559	33,769	36,132	38,662	41,368	44,264
Preferred Stock	124,568	See minority interest (I/S)	13,166	15,547	18,095	20,821	23,737	26,858	30,198	33,771
BV Equity	603,594	Constant	124,568	124,568	124,568	124,568	124,568	124,568	124,568	124,568
Liabilities & Equity	1,239,174	OBal + NI − Divs. − Repurch.	525,680	555,343	587,082	621,043	657,381	696,263	737,866	774,139
			1,180,987	1,249,261	1,322,314	1,400,481	1,484,119	1,573,612	1,669,370	1,763,587
Implied Assumptions										
Growth in BV			−13%	6%	6%	6%	6%	6%	6%	5%
Debt/(Debt + Equity)	26%		29%	30%	30%	30%	30%	30%	31%	31%
ROE	20.6%		23.2%	28.6%	29.0%	29.4%	29.8%	30.2%	30.6%	31.0%
Turnover	133.4%		144.9%	146.6%	148.2%	149.7%	151.1%	152.5%	153.8%	155.8%
DCF VALUATION										
Net Income to Common Stockholders			139,943	178,794	191,310	204,701	219,030	234,363	250,768	268,322
Cash Adjustments, Capital Exps, Financing										
Plus: Depreciation Exp			68,447	73,239	78,365	83,851	89,721	96,001	102,721	118,155
Plus: Deferred Tax			7,353	15,220	16,286	17,426	18,646	19,951	21,347	22,842
Less: Increase In Net Working Capital			55,811	(32,119)	(34,368)	(36,773)	(39,348)	(42,102)	(45,049)	(48,202)
Less: Capital Expenditures			(57,499)	(109,393)	(117,051)	(125,244)	(134,011)	(143,392)	(153,430)	(164,170)
Plus: Debt Issues			10,150	21,010	22,481	24,054	25,738	27,540	29,467	31,530
Less: Increased Cash			(8,573)	(17,745)	(18,988)	(20,317)	(21,739)	(23,261)	(24,889)	(26,631)
Plus: Preferred Stock Issues			0	0	0	0	0	0	0	0
Total Cash Adjs, Capital Exps, Financing			75,689	(49,789)	(53,274)	(57,004)	(60,994)	(65,263)	(69,832)	(66,477)
Adjust for Minority Interest		98.5% of total adjustment	74,554	(49,789)	(52,475)	(56,148)	(60,079)	(64,284)	(68,784)	(65,479)
Free Cash Flows to Equityholders			214,497	129,752	138,834	148,553	158,952	170,078	181,984	202,842
Growth in FCF				−39.5%	7.0%	7.0%	7.0%	7.0%	7.0%	11.5%
Cost of Equity Capital			16%							
Long-Term Growth Rate			7.0%							
Discount Factor			0.8621	0.7432	0.6407	0.5523	0.4761	0.4105	0.3538	0.3050
Present Value of Free Cash Flows			184,911	96,427	88,945	82,044	75,679	69,807	64,391	61,872
PV of Free Cash Flows to Date			184,911	281,338	370,283	542,328	528,006	597,814	662,205	724,077
Terminal Value										
							Forecast next year's FCF			217,041
							Terminal value = (FCF/(r − g)) x (PV factor)			735,590
							Annual merger cost savings ($60,000/year)			375,000
							Total present value of equity			1,834,667
							Adjust present value as if received thru year			1,981,440
							Number of shares			22,886
							Price			86.58

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Exhibit 1 Pro Forma Financial Statements and Valuation Using Lazard Assumptions

	1990		1991	1992	1993	1994	1995	1996	1997	1998
Pro Forma Assumptions										
Sales Growth	3.4%		3.5%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%
EBIT/Sales	12.9%		15.5%	15.5%	15.5%	15.5%	15.5%	15.5%	15.5%	15.5%
(1 − Average Tax Rate)	63%		63%	63%	63%	63%	63%	63%	63%	63%
= EBIT × (1 − t)/Sales	8.2%		9.8%	9.8%	9.8%	9.8%	9.8%	9.8%	9.8%	9.8%
(1 − Dividend Payout)	55%		55%	55%	55%	55%	55%	55%	55%	55%
Net Working Capital/Sales	16%		12.0%	12.0%	12.0%	12.0%	12.0%	12.0%	12.0%	12.0%
Depn Exp/Sales	3.6%		4.0%	4.0%	4.0%	4.0%	4.0%	4.0%	4.0%	4.3%
Capt Exp/Sales	5.7%		5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%
INCOME STATEMENT										
Net Sales	1,653,319	See sales growth assumption	1,711,185	1,830,968	1,959,136	2,096,275	2,243,015	2,400,026	2,568,028	2,747,789
EBIT	213,179	See EBIT/Sales assumption	265,234	283,800	303,666	324,923	347,667	372,004	398,044	425,907
Interest (net)	28,760	Debt/Interest grow at sales rate	29,767	31,850	34,080	36,465	39,018	41,749	44,672	47,799
EBT	184,419		235,467	251,950	269,586	288,457	308,649	330,255	353,373	378,109
Taxes	67,773	See tax rate assumption	87,123	93,221	99,747	106,729	114,200	122,194	130,748	139,900
Current Taxes			79,770	78,001	83,461	89,303	95,555	102,244	109,401	117,059
Deferred Taxes		Change in def tax liability	7,353	15,220	16,286	17,426	18,646	19,951	21,347	22,842
Discontinued Items	4,079									
Net Income Before Minority Interest			148,344	158,728	169,839	181,728	194,449	208,061	222,625	238,208
Minority Interest	—	1.5% of EBMI	2,225	2,381	2,548	2,726	2,917	3,121	3,339	3,573
Net Income	116,646		146,119	156,347	167,292	179,002	191,532	204,940	219,285	234,635
Preferred Dividend, net	6,176	Preferred stock constant	6,176	6,176	6,176	6,176	6,176	6,176	6,176	6,176
Net Income to Common Stockholders	114,549		139,943	150,171	161,116	172,826	185,356	198,764	213,109	228,459
CASH FLOW STATEMENT										
Cash from Operations										
Net Income	116,646		146,119	156,347	167,292	179,002	191,532	204,940	219,285	234,635
+ Depreciation	59,300	See assumptions	68,447	73,239	78,365	83,851	89,721	96,001	102,721	118,155
+ Deferred Taxes	16,933	Change in def tax liab	7,353	15,220	16,286	17,426	18,646	19,951	21,347	22,842
− Change in Working Capital	(7,580)	See Working Capt assumption	55,811	(14,374)	(15,380)	(16,457)	(17,609)	(18,841)	(20,160)	(21,571)
+ Other	(2,071)	See minority interest exp	2,225	2,381	2,548	2,726	2,917	3,121	3,339	3,573
Cash from Operations	183,228		279,955	232,813	249,110	266,548	285,206	305,171	326,533	357,634
Investment Cash Flow										
Capital Expenditures	(83,117)	See capt exp assumption	(85,559)	(91,548)	(97,957)	(104,814)	(112,151)	(120,001)	(128,401)	(137,389)
Asset Sales	198,531	Discontinued operations	36,681							
Increase in Leveraged Leases	(3,838)	Grows at rate of sales	(4,801)	(9,939)	(10,635)	(11,379)	(12,176)	(13,028)	(13,940)	(14,916)
Other Assets	—	Grows at rate of sales	(3,819)	(7,906)	(8,459)	(9,052)	(9,685)	(10,363)	(11,089)	(11,865)
	111,576		(57,499)	(109,393)	(117,051)	(125,244)	(134,011)	(143,392)	(153,430)	(164,170)
Financing										
New Debt Financing	(30,512)	Grows at rate of sales	9,218	19,080	20,416	21,845	23,374	25,010	26,761	28,634
Common Equity	6,224	Zero new financing	0	0	0	0	0	0	0	0
Preferred Equity	(34,916)	Zero new financing	0	0	0	0	0	0	0	0
Common Dividends	(50,128)	See payout assumption	(69,150)	(73,753)	(78,678)	(83,948)	(89,586)	(95,620)	(102,075)	(108,983)
Preferred Dividends	(9,956)	See dividends above	(6,176)	(6,176)	(6,176)	(6,176)	(6,176)	(6,176)	(6,176)	(6,176)
Other Liabilities	—		932	1,930	2,065	2,209	2,364	2,529	2,706	2,896
Financing Cash Flows	(109,332)		(65,177)	(58,919)	(62,374)	(66,070)	(70,024)	(74,256)	(78,784)	(83,629)
Foreign Exchange	3,069									
(Increase) Decrease in Cash	—		(8,573)	(17,745)	(18,988)	(20,317)	(21,739)	(23,261)	(24,889)	(26,631)
Free Cash Flow	188,541	Used to repurchase stock	148,707	46,756	50,699	54,918	59,432	64,262	69,431	83,204

PROSPECTIVE ANALYSIS
Application to the Schneider- Square D Case-Study

- Consensus: biased valuation:
  - ✓ Sales growth of 7% forever difficult to justify;
  - ✓ NWC/Sales too low;
  - ✓ Net CAPEX/Sales way too low (driving AT up even in the post-horizon period).

PROSPECTIVE FORECASTING

Application to the Schneider- Square D Case-Study

- A more reasonable set of key assumptions to be used in the prospective analysis to value Square D on a stand-alone basis?

  - ✓ Sales Growth of 3.5% next year and 5% per year thereafter;
  - ✓ EBIT/Sales of 15 to 16%;
  - ✓ Net Working Capital/Sales at 15%;
  - ✓ Projected Capital Expenditure/Sales of 5.2%;
  - ✓ Depreciation Expense/Sales at 4% between next year and 7 years from now and 4.3% thereafter.

PROSPECTIVE ANALYSIS

Application to the Schneider- Square D Case-Study

- For info:
  - ✓ Square D traded at $35 per share before any take-over rumors started to circulate;
  - ✓ Square D's share price jumped to $50 thereafter.
  - ✓ Square D's stand-alone valuation (Lazard Freres): $64 per share.
  - ✓ Square D's stand-alone valuation (revised assumptions): $34 per share.

PROSPECTIVE ANALYSIS
Application to the Schneider- Square D Case-Study

- So why would Lazard Freres wish to issue such a biased valuation?

DISCOUNTED PAY-OFFS' BASED VALUATIONS

DISCOUNTED PAY-OFFS' BASED VALUATIONS

		Approach	
		Cash-Flow	**Abnormal Income**
Focus	**Equity**	PVED, PVFCFE (PVND)	PVAE
	Net Operating Assets	PVFCF	PVANOPAT (PVAOI)

DISCOUNTED PAY-OFFS' BASED VALUATIONS

- Valuation methodologies:
  - ✓ Discounting the PV of the relevant flow up to infinity for cash-flow based approaches;
  - ✓ Adding the relevant current book value to the PV of the relevant flow up to infinity for accounting-based approaches;

- Valuation outcomes:
  - ✓ VE* (Intrinsic Value of Equity) if the focus is on equity;
  - ✓ EV* (Enterprise Value) if the focus is on net operating assets.

To derive the value of equity, one needs to add the intrinsic value of the firm's financial assets and substract the intrinsic value of other financial claims.

DISCOUNTED PAY-OFFS' BASED VALUATIONS

- Explicit Forecasting Period (EFP);
- Post-Horizon Period (PHP);
- The proportion of current intrinsic value generated in the PHP is much higher in cash-flow based than in accounting-based valuations:

  Positive NPV investments in the EFP:

  - ✓ depress FCFs in the EFP;
  - ✓ generate incremental AE or ANOPAT in the EFP.

FREE CASH-FLOW BASED VALUATION METHOD (PVFCF)

FREE CASH-FLOWS

- Ways to derive Free Cash Flows (FCF) to all claimholders:
  - ✓ FCF = NOPAT (OI after Tax) - ΔBVNOA
  - ✓ FCF = CFO - CAPEX
  - ✓ FCF = - ΔBVNFO + NFE + ND

    as:

    ND $\equiv$ FCFE = FCF + ΔBVNFO - NFE

FREE CASH-FLOWS

TABLE 19.4 Return on Common Equity (ROCE), Price-to-Book Ratios (P/B), and Price-Earnings (P/E) Ratios for Varying Levels of Levered Free Cash Flow (FCF), 1973–1990

FCF Group	*FCF/Price*	*ROCE*	*P/B*	*P/E*
1	87.1%	4.7%	0.78	17.4
2	23.4	9.2	0.91	10.3
3	10.8	12.1	1.10	9.6
4	6.3	13.7	1.32	10.4
5	2.2	14.6	1.55	11.6
6	−1.4	13.2	1.58	13.0
7	−6.1	12.6	1.49	12.8
8	−13.4	11.6	1.36	12.5
9	−25.9	9.9	1.14	12.2
10	−78.8	4.2	0.94	22.7

FREE CASH-FLOWS

- FCF is not a good indicator of profitability:
  - ✓ lower ROE associated with both low and high FCF;
  - ✓ ROE doesn’t vary too much over a wide range of positive and negative FCF.

FREE CASH-FLOWS

$$\frac{FCF_{t+1}}{BVNOA_t} = RNOA_{t+1} - \frac{(BVNOA_{t+1} - BVNOA_t)}{BVNOA_t}$$

- A firm's FCF captures both operating profitability and growth in net operating assets.

PRESENT VALUE OF FCF (PVFCF) METHOD

- The enterprise value of a firm can always be valued as:

$$EV_t^* = \sum_{i=1}^{+\infty} \frac{E_t(FCF_{t+i})}{(1+r)^i}$$

where:

r denotes the cost of the firm's capital.

PRESENT VALUE OF FCF (PVFCF) METHOD

- From the enterprise value, one obtains an intrinsic value of equity by:
  - ✓substracting the value of financial liabilities such as debt;
  - ✓adding the value of the non-operating (financial) assets held by the firm;

DRIVERS OF FCF-BASED VALUATIONS

- Assuming that FCF and NOPAT are expected to grow by g, with constant RNOA and IR, then:

$$g = RNOA * IR$$

with:

$$IR = \Delta BVNOA / NOPAT$$

- It follows that:

$$FCF \equiv NOPAT - \Delta BVNOA = NOPAT\left(1 - \frac{g}{RNOA}\right)$$

- And hence:

$$EV_t = \frac{FCF_{t+1}}{r - g} = \frac{NOPAT_{t+1}\left(1 - \frac{g}{RNOA}\right)}{r - g}$$

DRIVERS OF FCF-BASED VALUATIONS

- Future expected RNOA (or ARNOA);
- Future expected growth in the book value of net operating assets.

DRIVERS OF FCFE-BASED VALUATIONS

- Similarly, assuming that FCFE and CE (NP) are expected to grow by g, with constant ROE and PYR, then:

$$g = ROE\ (1\ - PYR)$$

with:

$$PYR\ \equiv ND/CE$$

- It follows that:

$$FCFE \equiv ND = CE * PYR = CE\left(1 - \frac{g}{ROE}\right)$$

- And:

$$VE_t = \frac{CE_{t+1}\left(1 - \frac{g}{ROE}\right)}{r_E - g}$$

DRIVERS OF FCFE-BASED VALUATIONS

- Future expected ROE (or AROE);
- Future expected growth in the book value of equity.

A PRACTICAL APPROACH TO VALUATION

A PRACTICAL APPROACH TO VALUATION

- Forecast future stylized income statements and balance sheets;

- Main drivers:
  - ✓ Sales Growth;
  - ✓ Gross Margins and Operating Expenses;
  - ✓ NOA of different types;
  - ✓ Net Financial Obligations;
  - ✓ ND (FCFE) paid as a cash dividend being the plug line in the spreadsheet.

A PRACTICAL APPROACH TO VALUATION

- Forecasting sales growth:
  - ✓ total market size and target market share over time;
  - ✓ Information provided by the firm;

- Sales growth tends to come down with time:
  - ✓ A common target growth rate is GDP growth;
  - ✓ Sales cannot grow faster than the economy in the long-term.

A PRACTICAL APPROACH TO VALUATION

- Forecasting growth margin (GM):
  - ✓ Select a target RNOA in the first year of the post-horizon period;
  - ✓ Keep in mind that RNOA (PM * NOAT) are mean-reverting over the long term;
  - ✓ Mean-reversion occurs through PM (and more specifically GM) as opposed to NOAT;
  - ✓ Select a target gross margin for the first year of the post-horizon period that is consistent with your target RNOA, given your forecasts of operating expenses and net operating assets;

A PRACTICAL APPROACH TO VALUATION

- Targets for RNOA mean-reversion:
  - ✓ Average level of RNOA in the industry?
  - ✓ Average level of RNOA in a comparable industry?
  - ✓ If accounting is unbiased:
    - r (WACC)?

A PRACTICAL APPROACH TO VALUATION

- Forecasting operating expenses:
  - ✓ Keep the ratio of operating expenses over revenue fairly constant except if it needs to be changed from year to year;
- Forecasting net operating assets:
  - ✓ Keep NOAT fairly constant except in the cases of short-term imbalances, expected changes in strategy or technology;
- Forecasting net financial obligations:
  - ✓ Keep leverage fairly constant if dealing with mature companies;

A PRACTICAL APPROACH TO VALUATION

- Estimate the cost of equity capital:
  - ✓ Use equity beta data available online;
  - ✓ Alternatively use regressions (market model);

- Risk-free rate: Yield on a 10-Yr+ government bond;

- Risk-premium:
  - ✓ US: 5%;
  - ✓ UK: 6%.

A PRACTICAL APPROACH TO VALUATION

- Deriving the cost of the firm's capital:

$$WACC = \frac{V_D}{V_D + V_E} r_D (1 - T_{CM}) + \frac{V_E}{V_D + V_E} r_E$$

with:

- T_{CM} denoting the marginal corporate tax rate;
- V_D denoting the market value of net debt;
- V_E denoting the market value of equity.

In most cases, NFO is a good proxy for V_D.